COURSE 2 · SECTION 9 · WORKED SOLUTION

# Field ppm by seat ring revision

Section 9 (videos S09.1 and S09.2) · Validation pass rate + field ppm

**DESIGN QUESTION**

Does the field data show a difference in seat ring wear between revs A, B and C, and how sure can we be about rev C?

## Steps

1. Open Section10_Start.pbix (or Engxtech_S09_2.pbix) > File > Save as > Engxtech_S09_Exercise.
2. Validation & field page: card Validation Pass Rate; ppm table with SeatRingRevs[SeatRingRev] and the five measures.
3. Right-click _Measures > New measure, three times (below). Add them to the ppm table as extra columns.
4. Compare rev C: Field ppm (matured) against ppm (all months, not valid). Keep the matured one on the page.

### DAX from the video

```
Builds[Matured]         = Builds[BuildMonth] + 120 <= DATE ( 2026, 9, 30 )
WarrantyClaims[Matured] = WarrantyClaims[BuildMonth] + 120 <= DATE ( 2026, 9, 30 )
WarrantyClaims[DaysToFailure]  = DATEDIFF ( WarrantyClaims[BuildMonth], WarrantyClaims[FailureDate], DAY )
WarrantyClaims[Within90]       = WarrantyClaims[DaysToFailure] <= 90
WarrantyClaims[IsSeatRingWear] = LEFT ( WarrantyClaims[FailureMode], 14 ) = "Seat ring wear"

Validation Pass Rate = DIVIDE ( [Lots Passed], [Lots Tested] )
Matured Units = CALCULATE ( SUM ( Builds[UnitsBuilt] ), Builds[Matured] = TRUE () )
Wear Failures 90d =
CALCULATE ( COUNTROWS ( WarrantyClaims ), WarrantyClaims[IsSeatRingWear] = TRUE (),
    WarrantyClaims[Within90] = TRUE (), WarrantyClaims[Matured] = TRUE () )
Field ppm = DIVIDE ( [Wear Failures 90d], [Matured Units] ) * 1000000
CI Low ppm =
VAR k = [Wear Failures 90d] VAR u = [Matured Units]
RETURN IF ( u > 0, IF ( k = 0, 0, CHISQ.INV ( 0.025, 2 * k ) / 2 ) / u * 1000000 )
CI High ppm =
VAR k = [Wear Failures 90d] VAR u = [Matured Units]
RETURN IF ( u > 0, CHISQ.INV ( 0.975, 2 * k + 2 ) / 2 / u * 1000000 )
```

### DAX new in this exercise

```
Wear Failures 90d (all months) =
CALCULATE ( COUNTROWS ( WarrantyClaims ), WarrantyClaims[IsSeatRingWear] = TRUE (), WarrantyClaims[Within90] = TRUE
() )
Units Built (all months) = SUM ( Builds[UnitsBuilt] )
ppm (all months, not valid) = DIVIDE ( [Wear Failures 90d (all months)], [Units Built (all months)] ) * 1000000
```

Why matured cohorts: a unit built in Aug 2026 has not yet had 90 days in service plus time for a claim to arrive. Counting it adds units but almost no failures, so the rate looks better than it is: rev C drops to 850 ppm (8/9407) if you include them. The interval is the exact Poisson interval on the failure count k, divided by the units. Rev B (81 failures) has a narrow interval; rev C (4 failures) is wide: it includes rev B's 2,892 and overlaps rev A, so the data cannot yet tell rev C apart from rev B or rev A. Say: rev C is an early signal; more matured months are needed.

## Expected result

| Check | Result |
|---|---|
| Validation test pass rate | **90.9 % (149/164)** |
| Matured build months | **Mar 2025 to Jun 2026** |
| Rev A | **2255 ppm (8/3548); 95 % CI 973-4443 ppm** |
| Rev B | **2892 ppm (81/28007); 95 % CI 2297-3595 ppm** |
| Rev C | **1710 ppm (4/2339); 95 % CI 466-4379 ppm** |
| Rev C incl. unmatured months (not a valid rate) | **850 ppm (8/9407)** |